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BMSVP: Blockchain-based message signing and verification protocol for security enhancement in smart grids

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Abstract

The transformation of traditional power grid systems towards smart grids is driven by factors such as the expansion of vehicle charging infrastructure, rapid technological advancements, demands for energy efficiency, as well as the surge in distributed energy generation. However, the smart grid system integrates with numerous internet of things (IoT) devices and legacy systems, most of which have inherent vulnerabilities or do not support current cryptographic protocols. In addition, the communication among the myriad of smart grid devices takes place over the public internet, further exposing the smart grids to security threats. Although numerous security techniques have been presented to mitigate these threats, the provision of perfect security at low overheads remains elusive. In this paper, we leverage blockchain, bilinear pairings, and elliptic curve cryptography to develop a provably secure protocol for smart grids. Its formal security is performed using the widely deployed Real or Random (RoR) model, while its semantic security analysis demonstrate its resilience against threats such as side-channeling, replays and privileged insiders. In addition, it is shown to offer authentication, untraceability, anonymity, and perfect key secrecy. In terms of performance evaluation, it is shown to reduce communication costs by 22.36% while maintaining fairly low computation overheads. There is therefore a need to further optimize this protocol to lessen the incurred computation overheads.

Keywords Attacks · Threats · Privacy · Smart grids · Authentication · Security · Efficiency

Introduction

The energy demands have continued to soar across the globe, necessitating the development of more sustainable techniques for generating, distributing and transmitting energy. Unfortunately, the traditional power grid systems are characterized with energy wastages, unidirectional information flow, rigid and inefficient configurations[1]. The smart grid (SG) utilizes advanced information communication technologies (ICTs), distributed data

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processing, and artificial intelligence to alleviate these challenges[2,3]. The SG integrates smart systems, renewable energy sources, as well as bidirectional flow of information and power to facilitate cooperation, real-time analysis, automation and enhance efficiency[4–6]. In addition, it offers demand-response management, enhanced energy efficiency, and reinforces dependable grid protection via self-healing techniques. A typical SG incorporates smart devices such as actuators, sensors, and smart meters that offer control, balancing, as well as accurate real-time monitoring of power systems[7]. It also integrates renewable sources of energy, monitoring systems, wireless modules, networks, and strong ICT infrastructures that improve grid efficiency, reliability, and access to scalable and distributed energy resources[8–10]. In this environment, the smart meters provide advanced metering, which permit more effective data collection. In contrast to traditional grids where metering data is recorded monthly or hourly, the SGs collect data every minute. As explained in[11], the SG is one of the notable application of Internet of Things (IoT), which incorporates ICT in power networks to enhance cost-effectiveness and efficiency in energy distribution, transmission, and generation.

As explained in[12], the transformation to smart grids is driven by a number of forces, such as rapid technological advancements, demands for energy efficiency, surge in distributed energy generation, as well as electric vehicle charging infrastructure expansion. In this regard, the SG is viewed as an interconnection of infrastructures that deploy digital intelligence in the management of power pricing, generation, consumption, and distribution. Its two way communication offers flexibility, reliability, and efficiency, which are critical in power operation and control[13,14]. The smart meters (SMs) in SGs regularly collect and transmit gas or electricity consumption information to the controller nodes. The high levels of automation and adaptability permit the SGs to effectively respond to evolving market settings and dynamic power demands. In addition, its two-way communication of information and electricity facilitate customer interaction, adaptive control and protection, as well as enabling the SGs to be self-healing.

It is clear that the SGs can potentially enhance dependability and efficiency of power systems. However, the increasing dependency on communication networks in the operation of SGs introduces cyber security risks[15] in these networks. For instance, the interconnected vital power plants may have vulnerabilities that can render these networks prone to attacks. These may include disclosure of sensitive data, power outages, and damages to the power network supporting infrastructure. In addition, SGs incorporate IoT to facilitate efficient and better operations of the energy systems, but these IoT devices have inherent vulnerabilities that can be inherited in SGs. Moreover, the smart grids have complicated architectures comprising of millions of smart devices, ICT networks and sensors spread across the entire power system[16]. In smart grid communication, existing standards such as IEC 61850-90-5 and IEEE C37.118.2 are deployed. However, these standards do not specify protected communication, secret key setup or node authentication. This leaves the power grid exposed to threats such as data tampering and unauthorized access. The integration of SGs with legacy power systems introduce numerous risks. This is because these legacy systems do not support current cryptographic protocols and security features[9]. Since these IoT devices are monitored and controlled by internet-based protocols, they are susceptible to malicious probing, hacking threats, and espionage[11].

The close and high degree of communication and coordination among these devices during power generation, distribution, transmission and consumption increases chances of cascaded failures as well as cyber-attacks. Worse still, smart grid systems deploy open communication networks in their communication[17]. This exposes them to threats such as message interception and impersonations. Although the smart meters simplify operations and enable precise monitoring, they introduce risks of loss of user privacy and data confidentiality[18]. In addition, the SMs measure and store huge volumes of data, which are then forwarded to utility companies, consumers or providers. Any successful interception of this private data can have serious consequences. For instance, the metering data may be deployed by attackers for user profiling. Similarly, attackers can use the captured data to determine the appliances installed on the user premises, keep track of users' lifestyle and monitor their power consumptions to infer whether they are at home or not. This way, attackers may plan to carry out burglary operations when users

are not at home. It is also possible for attackers to steal or tamper with the user data needed for proper functioning of the SGs, further threatening user privacy[19].

Other cyber threats in the SG environment include Denial of Service (DoS), malware, social engineering, replay, false data injection, and time delay attacks[14]. Basically, the goal of these SG attacks is to compromise integrity, confidentiality and availability. These may include interception of private consumer data, compromising reliability and stability of SGs energy distribution, power outages or explosions, as well as infrastructural damages[20,21]. Although the introduction of IoT enhances efficiency, reliability, and intelligence in smart grids, it renders the grid susceptible to threats such as man-in-the-middle (MitM), interception, alterations, and DoS[22–24]. All these threats can result into financial losses, operational disruptions, and even compromise to national security[25].

To curb the above security and privacy threats, authentication protocols have been developed to ensure protected interaction of various SG entities such as sensors, generation systems, consumers, smart devices and grid operators. In addition, there is need for proper SG entities identification, as well as the protection of exchanged data. In this regard, blockchain and IoT have been deployed to facilitate decentralized energy trading and distribution. However, these technologies introduce serious threats such as double spending, long latencies during transaction processing, forking, and single point failure owing to the centralized nature of smart grids[26]. Since the SMs are resource-limited, robust authentication schemes cannot be deployed in these devices[27]. Security protocols with high computation overheads hinder scalability in SGs since they impede the processing of numerous requests arriving at high rates. This is particularly true for SMs which are characterized with low storage capabilities, battery capacity and limited computational power. There is therefore need for a robust and yet efficient authentication protocol to secure messages exchanged over the public channels in the smart grid ecosystem.

Motivation

The ever increasing world populations have continued to push for higher energy demands globally. Therefore, there has been increased number of power consumers, distributors and connections among devices, homes and automobiles. This presents some challenges to the traditional power grid systems in terms of control and management of this largely centralized network. Effective management of these internet-based centralized systems requires advanced ICT and complex security protocols. However, any successful compromise of the central node results in the compromise of the entire network data. As such, the command node needs to be decentralized so as to mitigate this single point of failure and boost efficient operations. Unfortunately, this introduces complexities in security provision when the number of users and nodes connected to it surge. To enhance security in decentralized smart grids, blockchain technology has been deployed. Unfortunately, most of the current blockchain-based security techniques rarely offer user anonymity and perfect key secrecy. In addition, many of them are defenseless against threats such as replay, MitM, impersonation, ephemeral secret leakages, and privileged insiders[17]. Moreover, some of these current security solutions are based on sophisticated cryptosystems that require extensive processing and message exchanges. As such, they incur high communication and computation overheads, rendering them impractical for SG deployments.

Contributions

The conventional centralized smart grid systems are susceptible to single point of failure threats. In addition, the central nodes can be potentially compromised to bring forth data integrity challenges. As such, there has been a shift towards decentralized smart grid systems. Unfortunately, this transformation from centralized to decentralized systems introduces transparency challenges, extensive transaction overheads and long latencies that degrade network throughput and scalability. To this end, blockchain technology can be applied to enhance sustainability in distributed energy-sharing environment. In this regard, we make the following contributions:

- We leverage on Practical Byzantine Fault Tolerance (PBFT) consensus algorithm in the management of smart grid transactions such that transaction history is potted on all the blockchain's complete nodes. This helps preserve immutability and hence the security of smart grid transaction data.
- We deploy the bilinearity, computability and non-degeneracy characteristics of the bilinear maps to develop an authentication protocol that is semantically secure. Particularly, adversaries are unable to reverse these bilinear mappings, rendering our protocol extremely difficult to compromise. In addition, we incorporate elliptic curve cryptography primitives during message signing and verification. This adds another layer of complexity as the adversary must now solve the Elliptic Curve Discrete Logarithm Problem (ECDLP) in order to compromise our protocol.
- We perform elaborate formal security analysis using the RoR model. The results indicate that the shared key is provably robust. In addition, the extensive semantic security analysis show that our protocol mitigates most of the conventional smart grid security threats such as privileged insider, MitM, replays and impersonations. This is unlike most of the current smart grid authentication protocols which fail to effectively mitigate these security threats.
- Extensive comparative performance evaluations are performed. The results show that the proposed protocol results in a 22.36% reduction in communication costs at moderately low communication costs.

The remainder of this article is structured as follows: Sect. 2 describes some of the related works in the research domain, while Sect. 3 details the system model of the proposed scheme. On the other hand, Sect. 4 presents the security analysis of our protocol, while Sect. 5 discusses its performance evaluation. Towards the end of this article, Sect. 6 concludes the paper and gives some probable future research scopes.

Related works

The security threats in smart grids have attracted a lot of attention from researchers, leading to the development of numerous cryptographic solutions. For instance, security techniques based on physical unclonable functions (PUFs) have been proposed in[17,28–31], and[32] to avert cloning and physical threats. However, PUF-based security solutions have stability challenges. In addition, the protocol in[29] has design flaws that allow gateway nodes to gain access to secret challenge-response pairs allied to the PUFs[33]. This compromises anonymity and can be utilized to launch DoS that can disrupt SG network operations. Therefore, other secure authentication techniques are introduced in[34–37] and [38]. However, these schemes deploy the centralized architectures that are prone to single-point failure and hence are not ideal for secure user information storage[26]. To offer decentralized security, blockchain-based security solutions have been presented in[39–48], and[49]. However, majority of these schemes incur high computation and storage costs, with some lacking elaborate security analyses. Similarly, the anonymous authentication protocol developed in[50] requires high computation costs, hence unsuitable for smart meters[26]. Moreover, the energy trading schemes in[51,52] are elusive in terms of efficiency and scalability[26]. Based on Diffie-Hellman and digital signature, an authentication scheme is presented in[53]. Although it offers enhanced security functionalities, it is computationally intensive due to numerous point multiplications. Similarly, the ECC-based protocol in[54] incurs high communication and computation overheads and hence not ideal for low-capacity smart grid devices[55].

To address inefficiencies issues in some of the above protocols, a scheme based on Schnorr signatures is developed in[56]. Although Schnorr signatures are efficient and secure, this scheme is vulnerable to impersonation and MitM attacks[57]. Similarly, the elliptic curve cryptography (ECC)-based scheme in[58] is lightweight and hence ideal for deployment in smart meters. However, this scheme has incorrect ECC implementations and hence this flaw can be misused to compromise session key security. Although the ECC-based schemes in[59] and [60] address this problem, its reliance on absolutely trusted registration center limits its applications where such centers are lacking. Similarly, the scheme in[61] requires secure channels during registration; hence its applicability is curtailed

in environments devoid of such channels. To eliminate reliance on traditional certificates and offer authentication efficiency, certificateless cryptographic scheme is introduced in[33]. Although this scheme lowers key management overheads, it is prone to collision risks[14]. On the other hand, the mutual authentication scheme in[62] resists DoS, MitM, impersonation and replay attacks but at the expense of high storage and computational overheads[27,63]. Therefore, efficient authentication techniques are developed in[64] and [65]. However, the scheme developed in[64] is susceptible to impersonation attacks[14], while the protocol in[65] is prone to de-synchronization, traceability and stolen verifier threats[66]. Therefore, improved security techniques are presented in[67–70], and[71]. However, these authentication schemes have high communication overheads. In addition, their designs do not consider attack vectors such as session hijacking and side-channeling.

It is evident that numerous security frameworks have been developed or proposed for the smart grid networks. However, the discussion of the related works above has shown glaring research gaps in most of the current works that must be addressed. Such issues revolve around vulnerability to attacks and failure to support critical security functionalities. In addition, some of these current schemes are extensive in terms of communication and computation costs. The proposed protocol alleviates these challenges through robust security protection at lowest communication overheads and relatively low computation overheads.

The system model

In this section, we present the network model, as well the various critical phases of the proposed protocol.

Network model

The network model of the proposed protocol comprises of the registration authority (RA), the smart gas meter (SG_i), cloud server (CS), and the utility server (US_j), as shown in Fig. 1. Here, the RA is assumed to be sufficiently protected and has high computation power. In this network model, the RA performs the duties of SG_i, CS, and US_j registration as well as derivation of some cryptographic tokens required during the subsequent registration and authentication phase. On the other hand, the SG_i monitors the energy consumption at the customer premises, transmitting period reports to the US_j for appropriate decision making. On the other hand, cloud server CS is integrated to provide powerful computational and storage resources required to analyze the collected data and facilitate in relevant predictions regarding power demands and consumptions.

In terms of execution, our scheme executes system initialization, registration, message signing and validation, and blockchain formation. Table 1 presents the symbols used throughout this paper. Compared with other public key cryptosystems such as the Rivest-Shamir-Adleman (RSA), ECC offers the same level of security but reduced key sizes. As such, we deploy ECC in the proposed authentication protocol for enhanced efficiency.

System initialization

This phase is executed by the registration authority RA and involves the derivation of system parameters to be deployed in the registration and authentication. To accomplish this, the following steps are performed.

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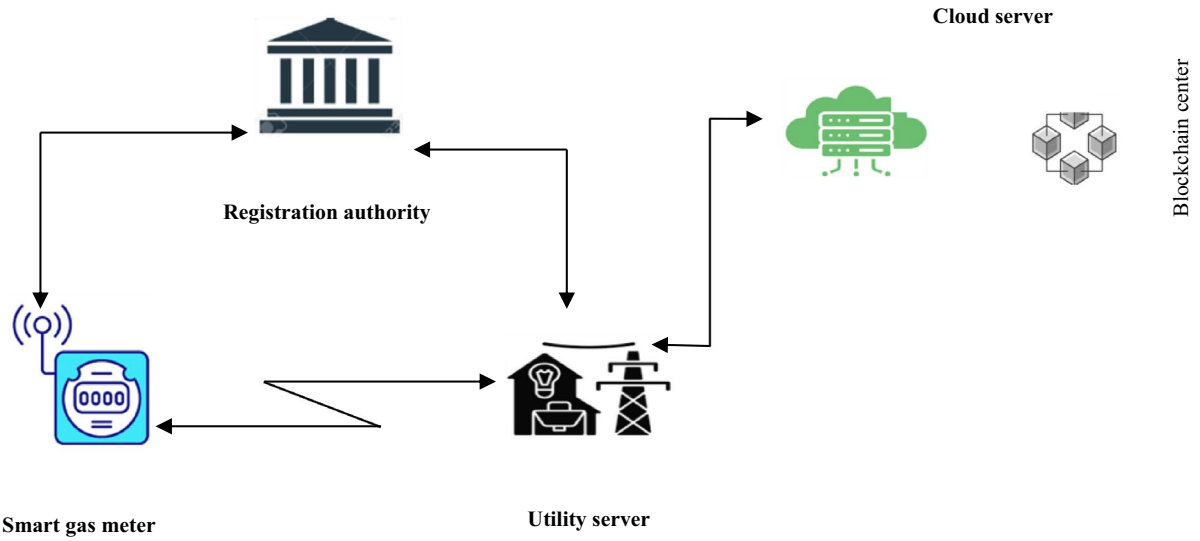


Fig. 1 Network model.

Table 1 Symbols.

Symbol	Description
RA	Registration authority
CS	Cloud server
US_j	Utility server j
SG_i	Smart grid i
P^r_{RA}	Private key for RA
r_k	Random nonce k
P^u_{RA}	Public key for RA
SID_i	Unique identity of SG_i
$USID_i$	Unique identity belonging to US_j
PID_i	Pseudo-identity of SG_i
PID_j	Pseudo-identity of US_j
P^r_{SG}	Secret key for SG_i
P^r_{US}	Secret key for US_j
P^r_{CS}	Secret key for CS
SK_{SU}	Shared key between SG_i and US_j
P^u_{SG}	Public key for SG_i
P^u_{US}	Public key for US_j
P^u_{CS}	Public key for CS
CE_k, CD_k	ECC-based encryption/decryption using key k
h (.)	Collision-resistant one-way hashing function
$\|\|$	Concatenation operation
S_p	Private ephemeral for SG_i
$\oplus$	XOR operation
C_k	Certificate k
Sig_k	Signature k
Δt	Maximum transmission latency
SE_k, SD_k	Symmetric encryption/decryption using key k

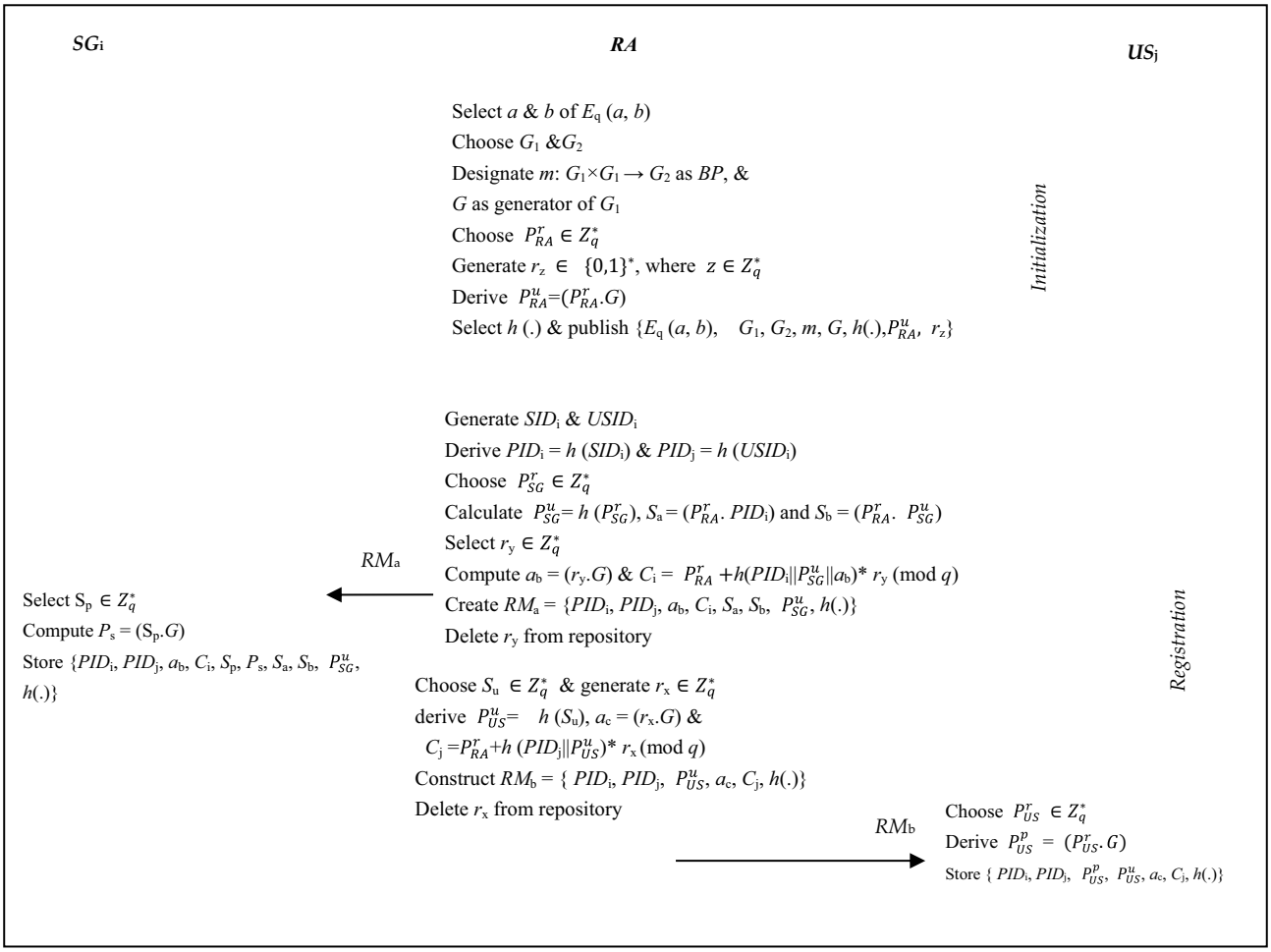


Fig. 2 System initialization and registration phase.

Step 1 The *RA* selects some two constants a and b of a non-singular elliptic curve E_q (a, b) whose form is $y^2 = x^3 + ax + b \ (modq)$ such that $4a^3 + 27b^2 \neq 0 \ (modq)$, where q is the order of this elliptic curve (*EC*). This is followed by the selection of G_1 and G_2 as additive and multiplicative groups, respectively, as shown in Fig. 2.

Step 2 The registration authority *RA* designates m and G as a bilinear mapping (*BP*) and generator of G_1, respectively. Here, the bilinear mapping m: $G_1 \times G_1 \rightarrow G_2$ exhibit all the bilinearity, computability and non-degeneracy features as described in[63].

Step 3 The private key of *RA* is stochastically choosen as $P^r_{RA} \in Z^*_q$ before generating random nonce $r_z \in \{0,1\}^*$, where $z \in Z^*_q$. Next, the *RA*'s corresponding public key is derived via the *EC* scalar point multiplication as $P^u_{RA} = (P^r_{RA}.G)$.

Step 4 The *RA* chooses some collision-resistant one-way hashing function h (.), followed by the publication of values $\{E_q (a, b), G_1, G_2, m, G, h(.), P^u_{RA}, r_z\}$. Algorithm 1 summarizes these system initialization procedures.

Registration

In this phase, all the smart gas meters and utility servers are registered at the TA. Algorithm 1 gives a summary

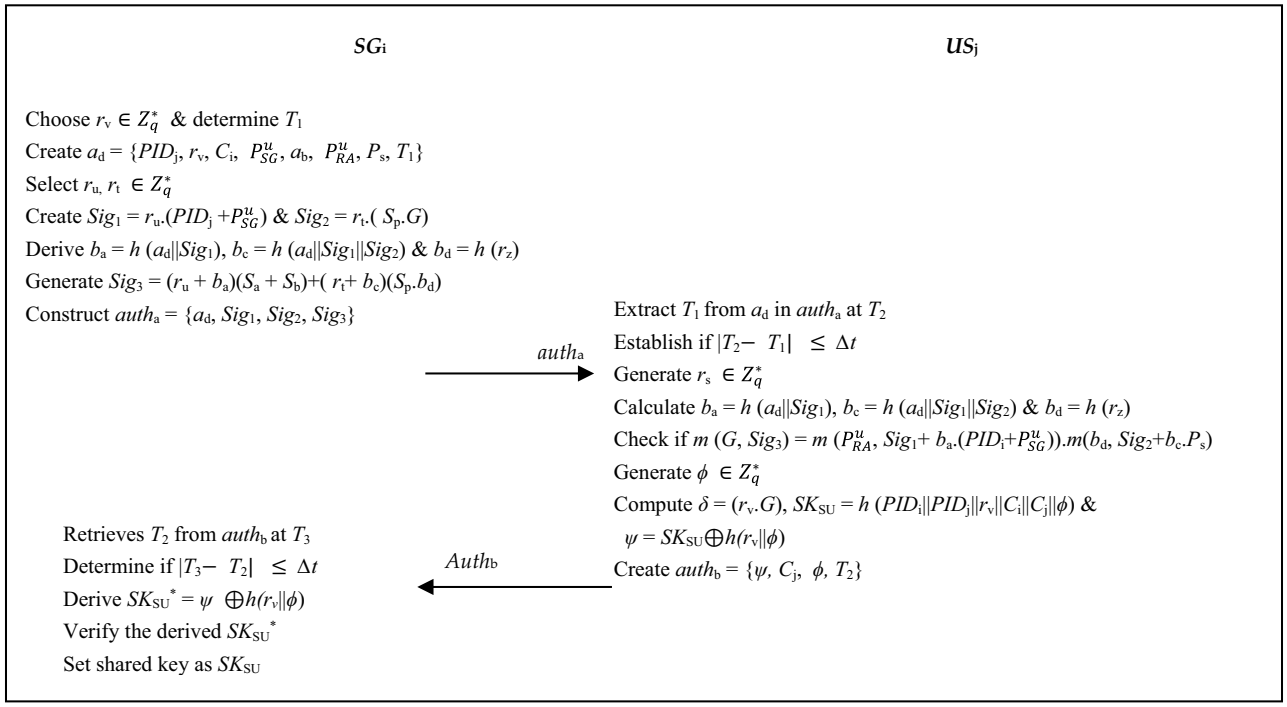


Fig. 3 Message signing and validation.

of these various steps carried out during the registration process.

Begin

**** *System initialization* *****

1) Choose a & b of $E_q(a, b)$
2) Selecet G_1 & G_2
3) Designate private key as $P_{RA}^r \in Z_q^*$
4) Generate $r_z \in \{0,1\}^*$
5) Derive public key as P_{RA}^u
6) Choose $h(.)$ & broadcast $\{E_q(a, b), G_1, G_2, m, G, h(.), P_{RA}^u, r_z\}$

**** *Smart gas registration* *****

7) Generate SID_i & derive PID_i
8) Designate as P_{SG}^r as the smart gas meter's secret key
9) Derive P_{SG}^u, S_a & S_b
10) Choose r_y
11) Calculate a_b & C_i
12) Construct RM_a
13) $RA \xrightarrow{RM_a} SG_i$
14) Delete r_y from RA's database
15) Choose $S_p \in Z_q^*$ & derive P_s
16) Store $\{PID_i, a_b, C_i, S_p, P_s, S_a, S_b, P_{SG}^u, h(.)\}$ in SG_i's memory

**** *Utility server registration* *****

17) Generate $USID_i$ & calculate PID_j
18) Choose $S_u \in Z_q^*$ & compute P_{US}^u
19) Generate $r_x \in Z_q^*$
20) Derive a_c & C_j
21) Compose RM_b
22) $RA \xrightarrow{RM_b} US_j$
23) Delete r_x from RA's database
24) Choose and designate $P_{US}^r \in Z_q^*$ as US_j' s private key
25) Compute P_{US}^p
26) Store $\{PID_j, P_{US}^p, P_{US}^u, a_c, C_j, h(.)\}$ in US_j' s repository

End

Algorithm 1 System initialization and registration.

Smart gas registration

In this phase, the registration authority selects the unique identity of smart gas meter *i*. In addition, it chooses the public key for this particular smart meter, as well as the partial secret key for smart gas meter *i*. The other computations performed in this phase are outlines in the steps below.

Step 1 The registration authority *RA* generates and desiginates SID_i as the unique identity of the smart gas meter SG_i. This is followed by the computation of the smart meter pseudo-identity $PID_i = h\ (SID_i)$, as shown in Fig. 2.

Step 2 The *RA* selects and designates $P_{SG}^r \in Z_q^*$ as the smart gas meter's secret key. Next, the corresponding public key is derived as $P_{SG}^u = h\ (P_{SG}^r)$. This is followed by the derivation of secre tokens $S_a = (P_{RA}^r.\ PID_i)$ and $S_b = (P_{RA}^r.\ P_{SG}^u)$.

Step 3 The registration authority *RA* stochastically selects nonce $r_y \in Z_q^*$ and derives $a_b = (r_y.G)$ as well certificate $C_i = P_{RA}^r + h(PID_i || P_{SG}^u || a_b)^* r_y \pmod q$.

Step 4 The *RA* creates registration message $RM_a = \{PID_i, PID_j, a_b, C_i, S_a, S_b, P_{SG}^u, h(.)\}$. Next, message RM_a is forwarded to the smart gas meter SG_i over protected commnication channels. Finally, the *RA* erases secret key r_y from its repository so as to thwart privileged insider threats.

Step 5 On getting message RM_a, the smart meter SG_i selects some secret ephemeral $S_p \in Z_q^*$. Next, it calculates the corresponding public ephemeral $P_s = (S_p.G)$. Finally, it stores $\{PID_i, PID_j, a_b, C_i, S_p, P_s, S_a, S_b, P_{SG}^u, h(.)\}$ in its memory.

Utility server registration

The ultimate goal of this phase is to register the utility server at the power generation company's premise to the *RA*. This is accomplished through the execution of the following steps.

Step 1 The registration authority *RA* generates $USID_i$ as the utility server's unique identity. Next, it calculates the corresponding pseudo-identity as $PID_j = h\ (USID_i)$, as evidenced in Fig. 2.

Step 2 The *RA* selects some stochastic key $S_u \in Z_q^*$ and computes its corresponding public key as $P_{US}^u = h\ (S_u)$. Next, it generates random nonce $r_x \in Z_q^*$ and calculates $a_c = (r_x.G)$ and certificate $C_j = P_{RA}^r + h\ (PID_j || P_{US}^u)^* r_x \pmod q$.

Step 3 The registration authority *RA* constructs registration message $RM_b = \{ PID_i, PID_j, P_{US}^u, a_c, C_j, h(.)\}$. It then forwards to the utility server US_j over secured communication channels. Meanwhile, the *RA* erases secret key r_x from its repository to mitigate any privileged insider threats.

Step 4 Upon receiving registration message RM_b, the utility server US_j selects $P_{US}^r \in Z_q^*$ and designates it as its private key. This is followed by the derivation of its corresponding public key as $P_{US}^p = (P_{US}^r.G)$. At the end, it stores values $\{PID_i, PID_j, P_{US}^p, P_{US}^u, a_c, C_j, h(.)\}$ in its repository.

Message signing and validation

In this phase, we deploy certificates and signatures to execute mutual authentication between the smart gas meter SG_i and utility server US_j. This is facilitated by the execution of the following steps, which are summarized in **Algorithm2**.

Step 1 The smart gas meter SG_i selects random nonce $r_v \in Z_q^*$ and determines the current timestamp T_1. Next, it creates value $a_d = \{PID_j, r_v, C_i, P_{SG}^u, a_b, P_{RA}^u, P_s, T_1\}$. Next, it chooses some two random nonces $r_u, r_t \in Z_q^*$. This is followed by the creation of signatures $Sig_1 = r_u.(PID_j + P_{SG}^u)$ and $Sig_2 = r_t.(S_p.G)$, as shown in Fig. 3.

Step 2 The SG_i calculates $b_a = h(a_d||Sig_1)$ and $b_c = h(a_d||Sig_1||Sig_2)$. Next, it deploys the public value r_z advertised by the registration authority RA to calculate $b_d = h(r_z)$. At last, it generates the signature on a_d as $Sig_3 = (r_u + b_a)(S_a + S_b) + (r_t + b_c)(S_p.b_d)$. Finally, it creates authentication message $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$ that is forwarded to the utility server US_j over open channels.

Step 3 Upon receiving message $auth_a$ at timestamp T_2, the US_j extracts timestamp T_1 from a_d and determines whether $|T_2 \quad T_1| \leq \Delta t$. If T_1 is invalid, message is flagged as replay and the session is halted. Otherwise, it generates nonce $r_s \in Z_q^*$ and computes $b_a = h(a_d||Sig_1)$ and $b_c = h(a_d||Sig_1||Sig_2)$.

Step 4 Using the public value r_z published by RA, the US_j derives $b_d = h(r_z)$. Next, it verifies the received signature Sig_3. This verification proceeds as follows.

$m(G, Sig_3) = m(P_{RA}^u, Sig_1 + b_a.(PID_i + P_{SG}^u)).m(b_d, Sig_2 + b_c.P_s)$ (1).

where $Sig_3 = (r_u + b_a).(S_a + S_b) + (r_t + b_c)(S_p.b_d)$, $Sig_1 = r_u.(PID_j + P_{SG}^u)$ and $Sig_2 = r_t.(S_p.G)$.

Therefore,

$m(G, Sig_3) = m(G, (r_u + b_a).(S_a + S_b) + (r_t + b_c)(S_p.b_d))$ (2).

$= m(G, (r_u(S_a + S_b) + b_a.(S_a + S_b)).\ m(G, (r_t.S_p.b_d + b_c.S_p.b_d)))$ (3).

$= m(G, (r_u(P_{RA}^r.PID_i + P_{RA}^r.P_{SG}^u) + b_a.(P_{RA}^r.PID_i + P_{RA}^r.P_{SG}^u)).$

$m(b_d, (r_t.S_p.G + b_c.S_p.G)))$ (4).

$= m(P_{RA}^r.G, (r_u(PID_i + .P_{SG}^u) + b_a.(PID_i + P_{SG}^u)).$

$m(b_d, (r_t.S_p.G + b_c.S_p.G + b_c.S_p.G))) = m(P_{RA}^u, Sig_1 + b_a.(PID_i + P_{SG}^u))$. (5)

$m(b_d, Sig_2 + b_c.P_s)$. (6)

since Eq. (6) is equivalent to Eq. (1), signature Sig_3 has been successfully validated. Therefore, the utility server has effectively authenticated the smart gas meter SG_i.

Step 7 The utility server US_j generates some stochastic secret token $\phi \in Z_q^*$. It then derives $\delta = (r_v.G)$, shared key $SK_{SU} = h(PID_i||PID_j||r_v||C_i||C_j||\phi)$, as well as $\psi = SK_{SU} \oplus h(r_v||\phi)$. Finally, it shares SK_{SU} with the smart gas meter SG_i in message $auth_b = \{\psi, C_j, \phi, T_2\}$ as shown in Fig. 3.

Step 8 After getting message $auth_b$ at timestamp T_3, the SG_i retrieves timestamp T_2 and checks whether $|T_3 \quad T_2| \leq \Delta t$. Basically, message $auth_b$ is flagged as replay and the session is terminated if T_2 is invalid. Otherwise, it derives $SK_{SU}^* = \psi \oplus h(r_v||\phi)$, verifies whether $SK_{SU}^* = h(PID_i||PID_j||r_v||C_i||C_j||\phi)$ prior to setting the shared key as SK_{SU}.

Blockchain formation for transactions management

This phase involves smart gas meter SG_i, utility server US_j, as well as cloud servers. Here, each cloud server possess the secret-public key pairs (P_{CS}^r, P_{CS}^u), which are based on the elliptic curve. The public key P_{CS}^u is given as $P_{CS}^u = P_{CS}^r.G$, where $P_{CS}^r \in Z_q^*$ is the private key. During the voting process, the cloud servers apply the Practical Byzantine Fault Tolerance (PBFT) consensus algorithm. The choice of PBFT is informed by its efficiency during consensus building and quick transaction finalization. In the verification process, a leader CSL is chosen among the peer to peer (P2P) CSs. This selected leader then sends the entire block to its peer nodes. After successful Bk verification, it is committed and added to the public blockchain. In this phase, we consider two forms

of transactions: the transactional data from smart gas meter SG_i towards the utility server US_j (Tx_{SG_i,US_j}); and the transactional data from the utility server US_j towards the smart gas meter SG_i (Tx_{US_j,SG_i}). Using the shared key SK_{SU}, the smart gas meter and the utility server can securely exchange data between themselves.

Transactional data $SG_i \rightarrow US_j$ Suppose that TC_{SG_i}, D_{SG_i}, and $Sig_{D_{SG_i}}$ are the creation time for a given transaction at SG_i, smart gas meter data, and signature created using the ECDSA algorithm. Here, $Sig_{D_{SG_i}} = ECDSA.Sig_{S_p}[PID_i, TC_{SG_i}, D_{SG_i}, P_s]$. Therefore, the transaction Tx_{SG_i,US_j} assumes the following format.

$$Tx_{SG_i,US_j} = \{\text{PIDi}, TC_{SG_i}, D_{SG_i}, \text{Ps}, Sig_{D_{SG_i}}\}$$

Transactional data $US_j \rightarrow SG_i$ Let us denote TC_{US_j}, D_{US_j}, and $Sig_{D_{US_j}}$ as the creation time for a given transaction at US_j, utility server data, and signature created using the ECDSA algorithm. Here, $Sig_{D_{US_j}} = ECDSA.Sig_{Pr_{US}}[PID_j, TC_{US_j}, D_{US_j}, P^u_{US}]$. As such, the transaction Tx_{US_j,SG_i} has the following formar.

$$Tx_{US_j,SG_i} = \{PIDj, TC_{US_j}, D_{US_j}, P^u_{US}, Sig_{D_{US_j}}\}$$

Block creation, verification and addition to the blockchain is accomplished through the execution of the following steps.

Step 1 Using the shared key SK_{SU}, the smart gas meter SG_i creates message $Bl_a = SE_{SK_{SU}}[Tx_{SG_i,US_j}]$. Next, this message is securely sent to the utility server US_j.

Step 2 On receiving message Bl_a, the utility server deploys the shared key SK_{SU} to decipher this message to obtain $\{Tx_{SG_i,US_j}\} = SD_{SK_{SU}}[SE_{SK_{SU}}[Tx_{SG_i,US_j}]]$. It then verifies pseudo-identity PID_i in transaction $Tx_{SG_i,US_j} = \{PID_i, TC_{SG_i}, D_{SG_i}, P_s, Sig_{D_{SG_i}}\}$. Meanwhile, it constructs message $Bl_b = SE_{SK_{SU}}[Tx_{US_j,SG_i}]$ that is forwarded to the smart gas meter SG_i.

Step 3 For a given pool of transactions $Tx_k \in \{Tx_{SG_i,US_j}; Tx_{US_j,SG_i}\}$, block Bk is formed from m_t transactions obtained from pool Tx_k. Next, the utility server US_j creates signature Sig_{Bk} based on these m_t transactions and the

ECDSA algorithm as $Sig_{\mathrm{Bk}} = ECDSA.Sig_{Pr_{US}}[\{Tx_{\mathrm{k}} | k=1, 2,\ldots. m_{\mathrm{t}}\}]$. Next, it composes message $Bl_{\mathrm{c}} = \{\{Tx_{\mathrm{k}} | k=1, 2,\ldots. m_{\mathrm{t}}\}, Sig_{\mathrm{Bk}}\}$ that is forwarded to the *CS*' blockchain.

Begin

1) Select $r_{\mathrm{v}} \in Z_q^*$ & determine T_1
2) Create a_{d}
3) Select $r_{\mathrm{u}}, r_{\mathrm{t}} \in Z_q^*$
4) Create signatures Sig_1 & Sig_2
5) Compute b_{a}, b_{c} & b_{d}
6) Creates signature Sig_3
7) Compose message $auth_{\mathrm{a}}$
8) $SG_{\mathrm{i}} \xrightarrow{auth_a} US_{\mathrm{j}}$
9) Receive $auth_{\mathrm{a}}$ at T_2
10) Extract T_1 from a_{d}
11) **If** $|T_2 - T_1| \geq \Delta t$ **then:**
12) Flag $auth_{\mathrm{a}}$ as replay & halt session
13) **Else:**
14) Generate $r_{\mathrm{s}} \in Z_q^*$
15) Calculate b_{a}, b_{c} & b_{d}
16) **If** Sig_3 is invalid **then:**
17) Abort session
18) **Else:**
19) Generate $\phi \in Z_q^*$
20) Compute δ, SK_{SU} & ψ
21) Create message $auth_{\mathrm{b}}$
22) $US_{\mathrm{j}} \xrightarrow{auth_b} SG_{\mathrm{i}}$
23) Receive $auth_{\mathrm{b}}$ at T_3
24) Extract T_2
25) **If** $|T_3 - T_2| \geq \Delta t$ **then:**
26) Flag $auth_{\mathrm{b}}$ as replay & stop session
27) **Else:**
28) Derive SK_{SU}
29) Set session key as SK_{SU}

End

Algorithm 2 Message signing and validation.

Step 4 On receiving message Bl_{c}, the cloud server *CS* constructs a partial block *PBk* whose header contains timestamp T_{Bk}, Merkle Tree Root MTR_{Bk}, and signer's public key P_{SG}^{u}. On the other hand, the block payload comprises of a catalog of m_{t} transactions ($\{Tx_{\mathrm{k}} | k=1, 2,\ldots. m_{\mathrm{t}}\}$), as well as the *ECDSA* signature on block *Bk* (Sig_{Bk}).

Step 5 The *CS* uploads *PBk* to the blockchain network for the purpose of verifying the entire block E_{Bk} whose header contains the block version Bk_{Ver}, previous block hash $Hash_{\mathrm{PrBk}}$, timestamp T_{Bk}, Merkle Tree Root MTR_{Bk}, and the signer's public key P_{SG}^{u}. On the other hand, the block payload comprises of m_{t} transactions ($Tx_{\mathrm{k}} | k=1, 2,\ldots. m_{\mathrm{t}}\}$), the *ECDSA* signature on block *Bk* (Sig_{Bk}), as well as the current block hash $Hash_{\mathrm{CBk}}$.

Algorithm 3 presents the PBFT-based consensus process during validation and addition in the blockchain network. We designate number of valid votes, a particular peer cloud server node, follower node, and number of defective *CS* nodes as Vv_{c}, CS_{k}, CS_{f}, and fn_{CS} respectively. In addition, we let $P_{CS_{\mathrm{f}}}^{u}$, Res_{Vf} and VS_{f} be the public key, voting response, and validation status for CS_{f}. Similarly, the public key and private key for CS_{L} are

represented as $P^{u}_{CS_L}$ and $P^{r}_{CS_L}$ respectively. Moreover, we designate the commitment message and P2P cloud server network as *CM* and *PCSN* respectively.

Security analysis

To demonstrate that the proposed protocol is provably secure, we carry out both formal and semantic security analyses against some of the predominant threat models in the smart grid environment. The detailed descriptions of these analyses are given in the sub-sections below.

Input: $\{P^{r}_{CS}, P^{u}_{CS}\}$, fn_{CS}, & E_{Bk}, where $E_{Bk} = \{Bk_{Ver}, Hash_{PrBk}, T_{Bk}, MTR_{Bk}, P^{u}_{US}, Sig_{Bk}, Hash_{CBk}, \{Tx_k | k = 1, 2, \ldots m_t\}\}$

Output: Block addition commitment

Begin

Step 1: The L generates nonce $r_q \in Z^{*}_{q}$, current timestamp T_c and a vote request Req_{Vo}

Step 2: **For** each CS_k, **Do**:

Leader CS_L derives $BV_a = CE_{P^{u}_{CS}}[r_q, T_c, Req_{Vo}]$

CS_L creates Bk validation request message $Req_{Va} = \{Bk, BV_a, T_c\}$

CS_L forwards Req_{Va} to CS_k over public channels

End for

Step 3: **For** each CS_f in CS peer-to-peer network **Do:**

Received message Req_{Va} at timestamp T_c^{*}

If $|T_c^{*} - T_c| \leq \Delta t$ **then:**

Using P^{r}_{CS}, decipher BV_a as $\{r_q^{*}, T_c^{*}, Req_{Vo}^{*}\} = CD_{P^{r}_{CS}}[BV_a]$

If $T_c^{*} = T_c$ **then:**

Validate $\{MTR_{Bk}, Sig_{Bk}, Hash_{CBk}\}$ on block E_{Bk}

If E_{Bk} validation is successful **then:**

Create vote response / E_{Bk} validation status message $Res_{Vo} = \{CE_{P^{u}_{CS_L}}[r_q^{*}, Res_{Vf}, VS_f]\}$

End if

End if

End if

End for

Step 4: Set $Vv_c = 0$

Step 5: **For** each Res_{Vo} received from CS_f **Do:**

Using $P^{r}_{CS_L}$, the CS_L deciphers Res_{Vo} as $[r_q^{*}, Res_{Vf}, VS_f] = CD_{P^{r}_{CS_L}}[Res_{Vo}]$

If $r_q^{*} = r_q$ and $\{[Res_{Vf}, VS_f] = \text{valid}\}$ **then:**

Set $Vv_c = Vv_c + 1$

End if

End for

Step 6: **If** $Vv_c \geq (2 * fn_{CS}) + 1$ **then:**

Add E_{Bk} to the blockchain

Broadcast *CM* to *PCSN*

End if

Algorithm 3 Consensus for *Bk* validation and addition.

Formal security analysis

In this section, we deploy the widely acknowledged Real-or-Random (RoR) model to demonstrate the robustness of the shared key $SK_{SU} = h\,(PID_i || PID_j || r_v || C_i || C_j || \phi)$. This model is deployed owing to its capabilities of offering rigorous, formal framework for validating the security of Authenticated Key Exchange (AKE) protocols. The various components associated with the deployed RoR model are described below.

Participants In our scheme, two entities are involved during the message signing. These entities are the smart gas meter SG_i, and the utility server US_j. We let $\pi^{I_1}_{SG_i}$ and $\pi^{I_2}_{US_i}$ represent the random oracles (instances) of SG_i and US_j, respectively.

Accepted state In the deployed RoR model, a random oracle π^I is in accepted state if it received the last verified transmitted message. Suppose that all the exchanged messages for both parties are sequentially concatenated. For a given current session, we refer this as the session identification (SID) of π^I.

Partnering Suppose that two instances π^{I_1} and π^{I_2} are both joint partners of each other, are in accepted states, share same SID, and have mutually verified each other. Then, these two random oracles are said to be partners to each other.

We denote a particular adversary as $\acute{\alpha}$ and assume that this attacker has access to the following queries.

Reveal (π^I): The attacker uses this query to disclose the shared key SK_{SU}. In the developed authentication algorithm, this key is derived by the US_j as $SK_{SU} = h\ (PID_i||PID_j||r_v||C_i||C_j||\phi)$. On the other hand, the SG_i computes this session key as $SK_{SU}{}^* = \psi \oplus h(r_v||\phi)$.

Execute ($\pi^{I_1}_{SG_i}$, $\pi^{I_2}_{US_i}$): Using this query, adversary $\acute{\alpha}$ is able to intercept and eavesdrop all the publicly exchanged messages between the smart gas meter SG_i, and the utility server US_j. These messages include $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$ and $auth_b = \{\psi, C_j, \phi, T_2\}$.

Test (π^I): In this query, attempts are made by adversary $\acute{\alpha}$ to solicit for shared key $SK_{SU} = h\,(PID_i||PID_j||r_v||C_i||C_j||\phi)$ from instance π^I. When faced with this query, the random oracle responds with stochastic outcome of a flipped fair coin, C.

Freshness On condition that shared key SK_{SU} has not been disclosed to any attacker using the *Reveal* (π^I) query, then instance $\pi^{I_1}_{SG_i}$ or $\pi^{I_2}_{US_i}$ is said to be fresh.

Hash function In the developed protocol, we model the collision-resistant one-way hashing function h (.) as a random oracle, $\mathfrak{H}$.

The security of our protocol is hinged in the difficulties of solving the Elliptic Curve Discrete Logarithm Problem (ECDLP) which can be mathematically stated as follows: Given two point $R, S \in E_q\,(a, b)$, the task in *ECDLP* is to find an integer $x \in Z_q^*$ such that S = $x.R$. Here, x is the discrete logarithm to the base S.

Semantic security Let us consider an adversary $\acute{\alpha}$ executing in polynomial time t and whose interest is to break the semantic security of our BMSVP scheme by deriving shared key SK_{SU}. Taking C and C^* as the legitimate and guessed bits respectively, and $Pr\,[\lambda]$ as the probability of a stochastic event λ, the advantage $Adv_{\acute{\alpha}}^{BMSVP}(t)$ of breaking the semantic security of our protocol is given as follows.

$$Adv_{\acute{\alpha}}^{BMSVP}(t) = |2\Pr\left[C = C^*\right] - 1| \tag{1}$$

To demonstrate the resilience of the proposed protocol, we formulate and proof Theorem 1 as follows.

Theorem 1 Suppose that hash query, advantage of breaking ECDLP in polynomial time t, number of hash queries, and the range space of h (.) are given as hsh, $Adv_{\acute{\alpha}}^{ECDLP}(t), hq_n$, and $|\mathbb{H}|$ respectively. Then, the attacker's advantage of computing shared key SK_{SU} in polynomial time t is given by:

$$Adv_{\acute{\alpha}}^{BMSVP}(t) \leq \frac{hq_n{}^2}{|\mathbb{H}|} + 2\left(Adv_{\acute{\alpha}}^{ECDLP}(t)\right) \tag{2}$$

Proof *We define three adversarial games* $Game_k^{\acute{\alpha}} | k = 0,1,2$. *In each of these games, we let* $Succ_{\acute{\alpha}}^{Game_k}$ *represent an event that adversary ά correctly guesses random bit C in game* $Game_k^{\acute{\alpha}}$. *Therefore, the advantage that ά wins game* $Game_k^{\acute{\alpha}}$ *is defined as,*

$$Adv_{\acute{\alpha},\ Game_k}^{BMSVP} = Pr\left[Succ_{\acute{\alpha}}^{Game_k}\right] \tag{3}$$

The specific details of each of the three games are illustrated below.

$\boldsymbol{Game_0^{\acute{\alpha}}}$: This is the initial game in which an adversary performs an actual attack against the proposed authentication protocol. Based on the semantic security definition of the proposed protocol given in Eqs. (1),

$$Adv_{\acute{\alpha}}^{BMSVP}(t) = |2\left(Adv_{\acute{\alpha},\ Game_0}^{BMSVP}\right) - 1| \tag{4}$$

$\boldsymbol{Game_1^{\acute{\alpha}}}$: This game represents an eavesdropping attack in which an attacker intercepts all the exchanged messages exchanged over the insecure communication channels. In our scheme, messages $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$ and $auth_b = \{\psi, C_j, \phi, T_2\}$ are exchanged during message signing and authentication. Therefore, the attacker performs the Execute ($\pi_{SG_i}^{I_1}$, $\pi_{US_i}^{I_2}$) query, followed by Reveal (π^I) and Test (π^I) in that order. The ultimate goal is to determine whether the obtained shared key $SK_{SU} = h(PID_i||PID_j||r_v||C_i||C_j||\phi)$ is the legitimate one or is just a random value. Here, $PID_i = h(SID_i)$, $C_i = P_{RA}^r + h(PID_i||P_{SG}^u||a_b)^* r_y \pmod q$, $PID_j = h(USID_i)$, and $C_j = P_{RA}^r + h(PID_j||P_{US}^u)^* r_x \pmod q$. Evidently, this shared key is based on both long terms secrets $\{PID_i, PID_j\}$ as well as short term secrets $\{r_v, \phi\}$. Since none of the long term and short term secrets are exchanged in messages $\{auth_a, auth_b\}$, they cannot be extracted by the adversary ά. As such, interception and eavesdropping have not increased adversary ά's probability of winning game $Game_1^{\acute{\alpha}}$. As such, $Game_1^{\acute{\alpha}}$ and $Game_0^{\acute{\alpha}}$ are indistinguishable and hence,

$$Adv_{\acute{\alpha},\ Game_1}^{BMSVP}(t) = Adv_{\acute{\alpha},\ Game_0}^{BMSVP}(t) \tag{5}$$

$\boldsymbol{Game_2^{\acute{\alpha}}}$: In this game, attempts are made to derive the shared key $SK_{SU} = h(PID_i||PID_j||r_v||C_i||C_j||\phi)$. To accomplish this, hash queries are simulated, followed by the solving of ECDLP. As already noted, the derivation of SK_{SU} requires both long terms secrets $\{PID_i, PID_j\}$ as well as short term secrets $\{r_v, \phi\}$. To obtain r_v from $\delta = (r_v.G)$, the adversary must solve the ECDLP in polynomial time t. The advantage that adversary ά has in solving this ECDLP is denoted as $Adv_{\acute{\alpha}}^{ECDLP}(t)$. On the other hand, the retrieval of ϕ from $SK_{SU} = h(PID_i||PID_j||r_v||C_i||C_j||\phi)$ is tantamount to reversing the one-way hashing function, h (.), which is computationally infeasible. To randomly derive shared key SK_{SU} that is an exact replica of the valid key is tatamount to finding a collision in the hash function. Therefore, devoid of simulating hsh queries and solving the ECDLP, $Game_2^{\acute{\alpha}}$ and $Game_1^{\acute{\alpha}}$ are indistinguishable. In this scenario, the application of the birthday paradox yields the following.

$$|Adv_{\acute{\alpha},\ Game_1}^{BMSVP}(t) - Adv_{\acute{\alpha},\ Game_2}^{BMSVP}(t)| \leq \frac{hq_n{}^2}{2|\mathbb{H}|} + Adv_{\acute{\alpha}}^{ECDLP}(t) \tag{6}$$

Having executed all queries devoid of winning any game, the only option left for adversary ά is to guess random bit C. This is then followed by the Reveal (π^I) and Test (π^I) queries. The ultimate goal is to determine whether the guessed bit is SK_{SU} or just any random value. Therefore,

$$Adv_{\acute{\alpha},\ Game_2}^{BMSVP}(t) = \frac{1}{2} \tag{7}$$

Based on Eq. (4), Eq. (5) and Eq. (7), we obtain the following.

$$\frac{1}{2}(Adv_{\acute{\alpha}}^{BMSVP}(t)) = |Adv_{\acute{\alpha},\ Game_0}^{BMSVP}(t) - \frac{1}{2}| = |Adv_{\acute{\alpha},\ Game_1}^{BMSVP}(t) - Adv_{\acute{\alpha},\ Game_2}^{BMSVP}(t) \tag{8}$$

According to Eq. (6) and *Eq. (8)*, we get,

$$\frac{1}{2}(Adv_{\acute{\alpha}}^{BMSVP}(t)) \leq \frac{hq_n^{\ 2}}{2|\mathbb{H}|} + Adv_{\acute{\alpha}}^{ECDLP}(t) \tag{9}$$

Multiplying both sides of Eq. (9) by factor 2 yields,

$$Adv_{\acute{\alpha}}^{BMSVP}(t) \leq \frac{hq_n^{\ 2}}{|\mathbb{H}|} + 2\left(Adv_{\acute{\alpha}}^{ECDLP}(t)\right) \tag{10}$$

This completes the proof and hence an adversary is unable to compromise the proposed protocol. Consequently, shared key SK_{SU} is semantically secure and unbreakable by adversary $\acute{\alpha}$.

Informal security analysis

In this section, we show that the proposed scheme is secure under all the attacker capabilities in the Canetti and Krawczyk (C-K) threat model. The attacker capabilities in this model are well described in[63].

Ephemeral leakage and session hijacking attacks

In these threats, we consider two scenarios during which the derived shared key $SK_{SU} = h\ (PID_i||PID_j||r_v||C_i||C_j||\phi)$ can be compromised. Here, $PID_i = h\ (SID_i)$, $C_i = P_{RA}^r + h(PID_i||P_{SG}^u||a_b)^* r_y \pmod q$, $PID_j = h\ (USID_i)$, and $C_j = P_{RA}^r + h\ (PID_j||P_{US}^u)^* r_x \pmod q$.

Scenario 1 Suppose that adversary $\acute{\alpha}$ has access to long term secrets $\{P_{US}^r, PID_i, PID_j\}$. The next goal is to derive shared key SK_{SU}. However, that attacker $\acute{\alpha}$ has no knowledge of short terms secrets $\{r_v, \phi\}$. As such, it is impossible to derive SK_{SU}.

Scenario 2 Let us assume that adversary $\acute{\alpha}$ has executed session hijacking and hence has knowledge of short term secrets $\{r_v, \phi\}$. Using these values, attempts are made to derive shared key SK_{SU}. However, devoid of long term secrets $\{P_{US}^r, PID_i, PID_j\}$, this derivation flops.

Eavesdropping and replay threats

In our scheme, two messages are exchanged during the message signing and verification. This message is $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$, and $auth_b = \{\psi, C_j, \phi, T_2\}$, where $a_d = \{PID_j, r_v, C_i, P_{SG}^u, a_b, P_{RA}^u, P_s, T_1\}$, $Sig_1 = r_u.(PID_j + P_{SG}^u)$, $Sig_2 = r_t.(S_p.G)$, $Sig_3 = (r_u + b_a)(S_a + S_b) + (r_t + b_c)(S_p.b_d)$ and $\psi = SK_{SU} \oplus h(r_v||\phi)$. Let us assume that adversary $\acute{\alpha}$ has eavesdropped the open channel and has obtained messages {autha, authb}. Afterwards, attempts are made to modify and forward these messages to unsuspecting utility server USj and smart gas meter SGi. However, on receiving message autha at timestamp T2, the USj extracts timestamp T1 from ad. It then establishes whether $|T_2 \quad T_1| \leq \Delta t$ such that the message is flagged as replay and the session is halted if T_1 is invalid. The same procedure is followed to determine the freshness of timestamp T_2 in message $auth_b$. In this way, any message replay attempts against our scheme are easily detected and dealt with.

Impersonation attacks

The aim of these attacks is to enable the adversary $\acute{\alpha}$ to masquerade as legitimate smart gas meter SG_i. Therefore, the attacker intercepts messages $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$ and $auth_b = \{\psi, C_j, \phi, T_2\}$. Here, $a_d = \{PID_j, r_v, C_i, P^u_{SG}, a_b, P^u_{RA}, P_s, T_1\}$, $Sig_1 = r_u.(PID_j + P^u_{SG})$, $Sig_2 = r_t.(S_p.G)$, $Sig_3 = (r_u + b_a)(S_a + S_b) + (r_t + b_c)(S_p.b_d)$ and $\psi = SK_{SU} \oplus h(rv||\phi)$. Thereafter, adversary $\acute{\alpha}$ tries to create bogus messages $auth_a{}^b = \{a_d{}^b, Sig_1{}^b, Sig_2{}^b, Sig_3{}^b\}$ and $auth_b{}^b = \{\psi^b, C_j{}^b, \phi^b, T_2{}^b\}$. However, devoid of correct random secrets $\{r_v, r_u, r_t\}$, no valid security parameter a_d and signatures $\{Sig_1, Sig_2, Sig_3\}$ can be computed. Similarly, valid random secrets $\{r_v, \phi\}$ are required for the derivation of ψ, as well as a valid timestamp T_1 for the creation of security token a_d. Moreover, the adversary must have access to long term secrets $\{S_p, S_a, S_b\}$. As such, it is computationally infeasible for adversary to impersonate any legitimate SG_i.

Man-in-the-middle threats

The ultimate objective of these threats is to intercept messages $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$ as well as $auth_b = \{\psi, C_j, \phi, T_2\}$, and then attempt to create bogus signatures $\{Sig_1, Sig_2, Sig_3\}$. In addition, attempts are made to compute value ψ. Here $Sig_1 = r_u.(PID_j + P^u_{SG})$, $Sig_2 = r_t.(S_p.G)$, $Sig_3 = (r_u + b_a)(S_a + S_b) + (r_t + b_c)(S_p.b_d)$, $\psi = SK_{SU} \oplus h(rv||\phi)$, $SK_{SU} = h(PID_i||PID_j||r_v||C_i||C_j||\phi)$, $PID_j = h(USID_i)$, $a_b = (r_y.G)$, $b_a = h(a_d||Sig_1)$, $a_d = \{PID_j, r_v, C_i, P^u_{SG}, a_b, P^u_{RA}, P_s, T_1\}$, $S_a = (P^r_{RA}.PID_i)$, $S_b = (P^r_{RA}.P^u_{SG})$, $b_c = h(a_d||Sig_1||Sig_2)$, and $b_d = h(r_z)$. Therefore, the creation of these signatures and ψ require knowledge of long term secrets $\{S_p, P^r_{SG}, P^r_{RA}, USID_i, PID_i, PID_j\}$ as well as short term secrets $\{r_t, r_u, r_v, r_y, r_z, \phi\}$. To obtain some of these parameters, the adversary must solve the *ECDLP*, which is computationally infeasible.

Physical capture, side-channeling and privileged insider attacks

The aim of these threats is to physically capture SG_i and US_j, then deploy power analysis techniques to obtain the values stored in memory, and attempt to calculate the shared key $SK_{SU} = h(PID_i||PID_j||r_v||C_i||C_j||\phi)$. During the registration process, the smart gas meter SG_i stores values $\{PID_i, PID_j, a_b, C_i, S_p, P_s, S_a, S_b, P^u_{SG}, h(.)\}$ in its memory. On the other hand, the utility server US_j stores $\{PID_j, PID_i, P^p_{US}, P^u_{US}, a_c, C_j, h(.)\}$ in its repository. Evidently, side-channeling can help attacker $\acute{\alpha}$ to obtain only $\{PID_i, PID_j\}$. However, $\acute{\alpha}$ still requires values $\{r_v, C_i, C_j, \phi\}$ in order to derive shared key SK_{SU}. In addition, the registration authority erases keys $\{r_y, r_x\}$ from its repository so as to thwart privileged insider threats. Therefore, our scheme is effectively protected against these threats. Moreover, the shared key derivation involves short-term secrets r_v and ϕ that are indepedently derived at the SG_i and US_j, respectively, without the the participation of the registration athority. Consequently, privileged insider within the registration authority have no access to these keying parameters.

Anonymity and unlinkability

In the proposed protocol, message $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$ and $auth_b = \{\psi, C_j, \phi, T_2\}$ are exchanged over the public channel. Here, $a_d = \{PID_j, r_v, C_i, P^u_{SG}, a_b, P^u_{RA}, P_s, T_1\}$, $Sig_1 = r_u.(PID_j + P^u_{SG})$, $Sig_2 = r_t.(S_p.G)$ and $Sig_3 = (r_u + b_a)(S_a + S_b) + (r_t + b_c)(S_p.b_d)$, $\psi = SK_{SU} \oplus h(r_v||\phi)$, $SK_{SU} = h(PID_i||PID_j||r_v||C_i||C_j||\phi)$, $PID_j = h(USID_i)$, $C_i = P^r_{RA} + h(PID_i||P^u_{SG}||a_b)^* r_y \pmod q$, $a_b = (r_y.G)$, $P^u_{RA} = (P^r_{RA}.G)$, $P_s = (S_p.G)$, $b_a = h(a_d||Sig_1)$, $b_c = h(a_d||Sig_1||Sig_2)$, $b_d = h(r_z)$, $S_a = (P^r_{RA}.PID_i)$ and $S_b = (P^r_{RA}.P^u_{SG})$. It is evident that these messages do not contain real identities $\{USID_i, SID_i\}$ and hence anonymous communication is attained. As such, our protocol provides identity hiding, which helps preserve anonymity. Although PID_j in a_d incorporates real identity of the utility server $USID_i$, it is in hashed format. In addition, signatures $\{Sig_1, Sig_2, Sig_3\}$ incorporate random nonces $\{r_u, r_t\}$, while ψ incorporates random secrets $\{r_v, \phi\}$. In this way, the developed authentication protocol adds stochasticity to the exchanged messages.

Therefore, it is infeasible for the adversary to associate any message to a particular session or network entity and hence unlinkability is preserved.

Mutual authentication

During message signing and verification, the utility server US_j and smart gas meter SG_i validate each other through signatures $\{Sig_1, Sig_2, Sig_3\}$. This is accomplished by checking whether $m\ (G, Sig_3) = m\ (P_{RA}^u, Sig_1 + b_a.(PID_i + P_{SG}^u)).m(b_d, Sig_2 + b_c.P_s)$, where $Sig_3 = (r_u + b_a).(S_a + S_b) + (r_t + b_c)(S_p.b_d)$, $Sig_1 = r_u.(PID_j + P_{SG}^u)$ and $Sig_2 = r_t.(S_p.G)$. On the other hand, the SG_i authenticates the shared key derived at the US_j by deriving $SK_{SU}^{*} = \psi \oplus h(r_v||\phi)$ and verifying whether $SK_{SU}^{**} = h\ (PID_i||PID_j||r_v||C_i||C_j||\phi)$. During blockchain formation, US_j and SG_i verify each other through pseudo-identity PID_i in transaction $Tx_{SG_i,US_j} = \{PID_i, TC_{SG_i}, D_{SG_i}, P_s, Sig_{D_{SG_i}}\}$.

DoS attacks

In this attack, the adversary tries to retransmit old but valid messages $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$ and $auth_b = \{\psi, C_j, \phi, T_2\}$ so as to overwhelm the utility server US_j with unnecessary processing. Here, $a_d = \{PID_j, r_v, C_i, P_{SG}^u, a_b, P_{RA}^u, P_s, T_1\}$, $\psi = SK_{SU} \oplus h(r_v||\phi)$, $SK_{SU} = h\ (PID_i||PID_j||r_v||C_i||C_j||\phi)$, $Sig_1 = r_u.(PID_j + P_{SG}^u)$, $Sig_2 = r_t.(S_p.G)$ and $Sig_3 = (r_u + b_a).(S_a + S_b) + (r_t + b_c)(S_p.b_d)$. Upon getting message $auth_a$ at timestamp T_2, the US_j retrieves T_1 from a_d and establishes whether $|T_2 \quad T_1| \leq \Delta t$. Similar procedures are followed to establish the freshness of timestamp T_2 in message $auth_b$. Basically, the session is aborted if timestamps $\{T_1, T_2\}$ are invalid. In this way, unnecessary processing of $auth_a$ and $auth_b$ is avoided at the onset, hence saving the server's computational resources. Apart from initial timestamp checks, the US_j performs signature verification prior to deriving additional values $\{\phi, \delta, SK_{SU}, \psi\}$. Similarly, the SG_i derives and validates shared key prior to deploying it for encryptions.

Forward and backward secrecy

In authentication protocols, forward key secrecy ensures that past communications are safe in the event that current or future session keys are compromised. Therefore, an adversary with the current session key is unable to decrypt the enciphered messages exchanged in past communications. On the other hand, backward key secrecy offers protection to future communications in the event that session keys are compromised. In the proposed authentication protocol, the shared key is derived as $SK_{SU} = h\ (PID_i||PID_j||r_v||C_i||C_j||\phi)$. Here, $PID_i = h\ (SID_i)$, $C_i = P_{RA}^r + h(PID_i||P_{SG}^u||a_b)^* r_y \pmod q$, $PID_j = h\ (USID_i)$, and $C_j = P_{RA}^r + h\ (PID_j||P_{US}^u)^* r_x \pmod q$. The incorporation of random nonces $\{r_v, r_y, r_x\}$ renders this shared key quite different for each session. This means that different random nonces are generated for each communication session. As such, adversaries in possession of the current shared key SK_{SU} is unable to utilize it to derive shared keys deployed in past and future communication sessions. Consequently, even if long-term credentials $\{PID_i, PID_j\}$ are compromised after completion of a session, the attacker still needs valid short-term secrets such as random nonces to accurately derive previously established session keys.

Scalability

In most of the authentication protocols, the registration authority *RA* is involved during the message signing and verification, as well as during the mutual authentication procedures. This *RA* can therefore present some performance bottleneck as the number of smart devices surge within the network. This is due to the increased processing that must now be accomplished at the *RA*. To address this problem and preserve scalability, the utility server US_j and the smart gas meter SG_i directly execute message signing and verification devoid of the involvement of the *RA*.

Identity guessing attacks

Suppose that the adversary is interested in extracting or guessing the real identities $\{SID_i, USID_i\}$ from the publicly exchanged messages. In our authentication protocol, these messages include $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$, and $auth_b = \{\psi, C_j, \phi, T_2\}$. Here, $a_d = \{PID_j, r_v, C_i, P^u_{SG}, a_b, P^u_{RA}, P_s, T_1\}$, $Sig_1 = r_u.(PID_j + P^u_{SG})$, $Sig_2 = r_t.(S_p.G)$, $Sig_3 = (r_u + b_a)(S_a + S_b) + (r_t + b_c)(S_p.b_d)$, $\psi = SK_{SU} \oplus h(r_v||\phi)$, $SK_{SU} = h(PID_i||PID_j||r_v||C_i||C_j||\phi)$, $PID_j = h(USID_i)$, $C_j = P^r_{RA} + h(PID_j||P^u_{US})^* r_x \pmod q$, $PID_i = h(SID_i)$, $C_i = P^r_{RA} + h(PID_i||P^u_{SG}||a_b)^* r_y \pmod q$, $P^u_{SG} = h(P^r_{SG})$, $a_b = (r_y.G)$, $P^u_{RA} = (P^r_{RA}.G)$ and $P_s = (S_p.G)$. It is clear that the extraction of real identities in our protocol involves either reversing the collision-resistant one-way hashing function, or solving ECDLP. Since these operations are both computationally infeasible, the proposed protocol effectively mitigates identity guessing attacks.

Performance evaluation

In this section, we present the comparative evaluation of the proposed protocol against other related schemes. This evaluation is in terms of the incurred computation and communication overheads. These two performance measures are deployed here due to their prevalence in the comparative evaluation of other related authentication protocols. In addition, we compare the supported functionalities in our scheme against those suported by its peers. Moreover, we also describe the implementation details of the blockchain-based transactions management.

Computation overheads

In this sub-section, we deploy the cryptographic operations performed durig the message signing and verification to compute the execution time of the proposed scheme. Let us designate GIFT-COFB operation, PUF, map to point hash, ELGamal public key encryption, Pollards lambda method, scalar point multiplications, fuzzy extraction, ELGamal public key decryption, one-way hashing, modular exponentiation, bilinear pairings, string-to-point hash, point additions, as T_G, T_{puf}, T_{mtp}, T_{pe}, T_{pm}, T_{em}, T_{fe}, T_{pd}, T_h, T_{me}, T_{bp}, T_{sth}, and T_{ea} respectively. The server and gateway in the proposed protocol are represented by an HP ZBook machine equipped with a 4.6 GHz Intel Core i7-9850 H and 32GB of RAM, running on Ubuntu 22.04 LTS. On the other hand, the smart device or user device is represented by Raspberry Pi 3 Model B+ with a 64-bit ARM Cortex-A53. In these environments, Table 2 presents the execution durations.

During the message signing and verification, the smart gas meter SG_i performs $3T_h + 3T_{em} + 3T_{ea}$ operations. On the other hand, the utility server performs $5T_h + 5T_{em} + 4T_{ea} + 3T_{bp}$ operations. Therefore, the total computation cost

Table 2 Cryptographic execution duration.

Scheme	Cryptographic operations	
	Smart device/user	Server/Gateway
T_G	7.782	1.29
$\boldsymbol{T_{puf}}$	0.515	0.175
$\boldsymbol{T_{mtp}}$	8.736	2.85
$\boldsymbol{T_{pe}}$	17.821	4.97
$\boldsymbol{T_{pm}}$	1.835	0.645
$\boldsymbol{T_{em}}$	13.253	3.26
$\boldsymbol{T_{pd}}$	4.862	1.98
$\boldsymbol{T_h}$	0.151	0.035
$\boldsymbol{T_{me}}$	16.146	5.72
$\boldsymbol{T_{bp}}$	27.168	9.72
$\boldsymbol{T_{sth}}$	32.151	15.68
$\boldsymbol{T_{ea}}$	1.287	0.391
$\boldsymbol{T_{fe}}$	13.253	3.26

Table 3 Computation overheads.

Scheme	Cryptographic operations		Total (ms)
	Smart device/user	Server/Gateway	
[53]	$19T_{em} \approx 251.807$	$18T_{em} \approx 58.68$	≈ 310.487
[67]	$7T_h + 8T_{em} \approx 107.081$	$5T_h + 11T_{em} + 4T_{bp} \approx 74.88$	≈ 181.961
[68]	$2T_h + 4T_{em} + T_{ea} \approx 54.601$	$4T_h + 8T_{em} + T_{ea} \approx 26.61$	≈ 81.211
[69]	$8T_h + 3T_{em} + T_{fe} \approx 54.22$	$10T_h + 4T_{em} \approx 13.39$	≈ 67.61
[70]	$T_{pe} + T_{mtp} + T_{me} + 2T_{em} + T_G \approx 76.991$	$T_{pe} + T_{mtp} + T_{me} + 5T_{em} + 5T_G + T_{pd} \approx 38.27$	≈ 115.261
[71]	$4T_{em} + 4T_h \approx 53.616$	$12T_{em} + 9T_h \approx 39.435$	≈ 93.051
Proposed	$3T_h + 3T_{em} + 3T_{ea} \approx 44.073$	$5T_h + 5T_{em} + 4T_{ea} + 3T_{bp} \approx 47.195$	≈ 91.268

Table 4 Communication costs.

Scheme	No. of messages	Total (bits)
[53]	2	2816
[67]	4	5184
[68]	4	4009
[69]	4	2976
[70]	2	2432
[71]	4	3520
Proposed	2	1888

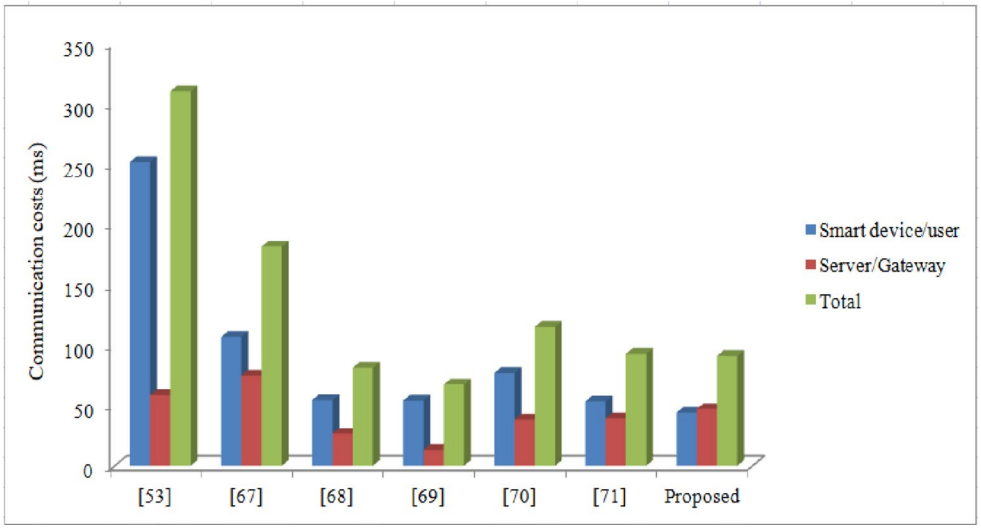


Fig. 4 Computation overheads comparisons.

is $8T_h + 8T_{em} + 7T_{ea} + 3T_{bp}$ as shown in Table 3. Using the cryptographic runtimes in Table 2, the computation overheads at the smart device/user, as well as server/gateway for various protocols are derived as shown in Table 3.

As shown in Fig. 4, the scheme in[53] incurs the highest computation costs of 251.807 ms at the smart device/user side. This is followed by the schemes developed in[67–70] and [71] with 107.081 ms, 76.991 ms, 54.601 ms, 54.22 ms and 53.616 ms computation overheads, respectively. On the other hand, the proposed authentication protocol incurs the least computation costs of only 44.073 ms at the smart device/user side. At the server/gateway side, the scheme in[67] incured the highest computation costs of 74.88 ms, while the protocol developed in[69] incured the least computation overheads of only 13.39 ms. In overall, the protocol developed in[53] took the longest execution time of 310.487 ms, while the scheme in[69] took the shortest time of only 67.61 ms to execute. Although the protocol in[69] is generally the most efficient computationally, it has relatively high communication overheads as shown in Table 4. In addition, it fails to defend against various attacks such as session hijacking, Dos and side-channeling, as illustrated in Table 5.

Table 5 Supported functionality.

	[53]	[67]	[68]	[69]	[70]	[71]	[Ours]
Functionality							
Formal analysis	√	√	×	√	√	√	√
Authentication	√	√	√	√	√	√	√
Anonymity	×	√	√	√	√	√	√
Untraceability	×	√	√	×	√	√	√
Forward secrecy	√	√	×	√	×	×	√
Backward secrecy	√	√	×	√	×	×	√
Scalability	×	×	×	×	×	×	√
Resilience against							
Ephemeral leakage	×	√	×	√	×	√	√
Session hijacking	×	×	×	×	×	×	√
Eavesdropping	×	√	×	√	√	×	√
Replay	√	√	√	√	√	√	√
Impersonation	√	√	×	√	×	√	√
MitM	√	√	×	√	√	√	√
Physical capture	×	√	×	√	×	√	√
Side-channeling	×	×	×	×	×	×	√
Privileged insider	×	√	×	√	√	×	√
DoS	×	√	×	×	×	×	√
Offline guessing	×	√	×	√	×	√	√

√: Feature present; ×: feature absent or not considered

Although the protocols in [68 and 69] incur slightly low computation overheads, they consume high bandwidths. Owing to their extensive communication overheads. In addition, they are not evaluated against critical attacks such as side-channeling and session hijacking. However, our protocol mitigates these attacks and supports all these features at relatively low computation overheads. In the smart grid ecosystem, the smart/user devices are resource-constrained. Therefore, the proposed authentication protocol is the most ideal for deployement in these devices. On the other hand, the gateways and servers have huge computational resources and hence can manage the relatively intensive operations executed in our scheme, such as bilinear pairing operations.

Communication overheads

In this sub-section, we utilize the size of the messages transmitted during message signing and verification to derive the cumulative communication costs of our scheme. During these procedures, messages $auth_a = \{a_d, Sig_1, Sig_2, Sig_3\}$ and $auth_b = \{\psi, C_j, \phi, T_2\}$ are exchanged. We take the sizes of ECC point, timestamp, hash value, identifiers, and random nonces to be 160 bits, 32 bits, 160 bits, 64 bits, and 128 bits respectively. Therefore, $auth_a$ = {64+128+160+128+160+128+160+32+160+160+160=1440 bits}, and $auth_b$ = {160+160+128=448 bits}. Consequently, the cumulative communication cost of our scheme is 1888 bits as shown in Table 4.

As shown in Fig. 5, the scheme in[67] exchanges 4 messages with total communication cost of 5184 bits, which is the highest. This is followed by the protocols in[53,68,69,71] and [70] with communication costs of 4009 bits, 3520 bits, 2976 bits, 2816 bits, and 2432 bits respectively.

On the other hand, our scheme incurs a communication overhead of only 1888 bits, which is the lowest. Since network bandwidth is a scarce resource, it should be utilized in an efficient manner. Therefore, the protocol developed in[67] is the most wasteful, while the proposed scheme is the most efficient. Using the protocol presented in[70] as the baseline, our protocol reduced the communication overhead by 22.36%.

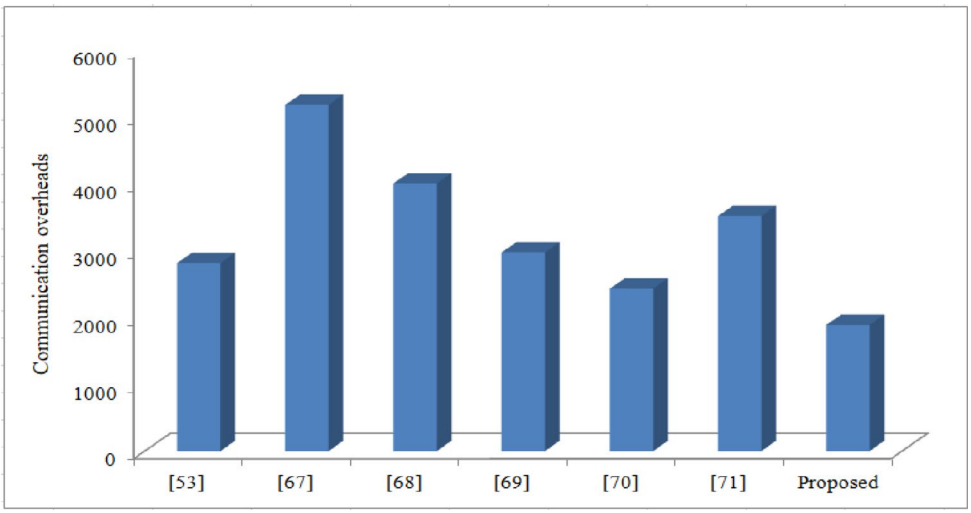


Fig. 5 Communication overheads comparisons.

Supported functionalities

To demonstrate that our scheme is secure against typical smart grid attacks, we compare its supported security features and attack resilience against its related peers as shown in Table 4. As shown in Table 5, the schemes developed in[68] supports only 4 functionalities, hence is the most vulnerable. This is followed by the protocols in[53,69–71] and [67] which support 7, 8, 10, 13, and 15 functionalities, respectively. On the contrary, the proposed protocol supports all the 18 functionalities. As such, our scheme offers robust security protection at the lowest communication costs, relatively low computation costs. It therefore strikes a good balance between performance and security. This projects it as the most suitable for deployment in smart grids environment, where majority of the smart devices are characterized with resource limitations in terms of computation and communication capabilities.

Blockchain implementation

To realize practical implementation of the proposed BMSVP, we deployed the Hyperledger Sawtooth (HS). This is an enterprise blockchain platform that is well suited for applications that are distributed in nature. The validator nodes in this HS network validate transactions and add blocks to the blockchain.

It therefore offers consensus using the PBFT algorithm. Our scheme is simulated using the P2P servers that act as validator nodes. The configuration of these servers is as described in sub-Sect. 5.1 above. To investigate the effects of number of mined blocks (B) on the computation costs, we increment the number of mined blocks but keep the number of transactions per block constant (50 transactions per block). As shown in Fig. 6, there is a near linear surge in execution time (from an initial value of 8.527 s to 50.292 s) as the number of mined blocks increases from an initial value of 10 to 50.

In the second scenario, the number of mined blocks is kept constant, while the number of transactions per block (T) is varied. The output in Fig. 7 shows that as the number of mined transactions per block is incremented from 25 to 100, there is a corresponding increase in the execution time from an initial value of 15.289 s to 35.289 s.

These graphs on the effects of number of mined blocks on the computation costs, as well as the influence of the number of transactions per block on execution time are significant in that they help visualize the behaviour of network load during transaction management. This helps in the optimization of the network computational resources so as to achieve enhanced performance for various loads.

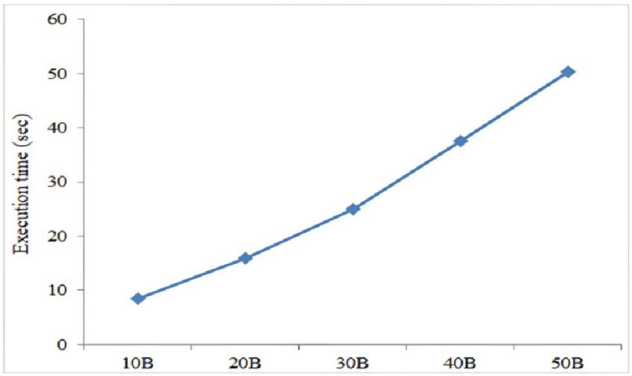


Fig. 6 Number of mined blocks.

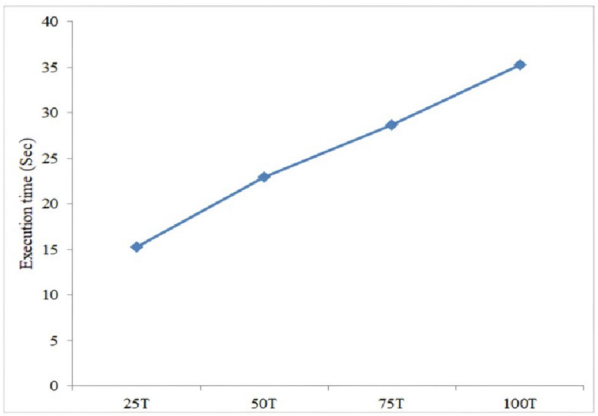


Fig. 7 Transactions per block.

Conclusion

The smart grid employs advanced information communication technologies, distributed data processing, and artificial intelligence to address critical challenges in traditional grid systems, such as inefficient configurations and energy wastages. However, the amalgamation of numerous IoT devices with their numerous vulnerabilities, as well as integration with conventional grid systems renders these smart grids susceptible to numerous security threats. In this regard, a myriad of solutions have been presented over the recent past. However, there are notable challenges in most of these schemes and hence the provision of perfect security at low costs is still a mirage. To bridge these gaps, we have developed a robust message signing and verification technique that is demonstrated to be provably secure under the RoR model. In addition, it mitigates typical smart grid threats such as ephemeral leakage, session hijacking and eavesdropping. Moreover, it offers salient security functionalities such as forward secrecy. From the performance perspective, it incurs the lowest communication overheads and slightly lower computation costs. Future research scopes in this domain revolve around further reductions in the attained computation costs.

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Data availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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